College of Arts and Science

School of Computing

STIS 3143 GROUP A

Social Media Data Analytics

Assignment 2

Group 1

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Name	Matric Number
Lee Shi Hang	278841
Ho Sheng Yu, Bryan	279455
Jiji Ong	279806
Lim Yong Kang	279868

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Submission to: Ts. Dr. Juhaida Binti Abu Bakar

1.0 INTRODUCTION

In this age of digitalization, the emergence of social media has fundamentally changed how individuals exchange information and communicate. While there are no doubt many advantages in social media, there is also a darker side, which is the surge in hate speech. Hate speech is any verbal or written expression that is harmful or threatening to a person or group because of that person's or group's unique characteristic (Udanor & Anyanwu, 2019).

Like many other nations, Malaysia has seen an alarming increase in the dissemination of hate speech on social media in recent years. Examples of hate speech include threats, defamation, which includes libel and slander, sentiments or attitudes of hatred, obscenity, and other such expressions on social media. Hate speech is also defined by the level of attention paid to the substance and tone of the communication (Udanor & Anyanwu, 2019). It is safe to say that hate speech poses a serious danger to religious peace, national unity, and social cohesiveness. Therefore, it is essential to comprehend the nature of hate speech on social media in Malaysia and look into practical solutions to address this expanding problem.

The distinctive socio-political environment and the quick development of internet connectivity have increased Malaysia's vulnerability to the danger of hate speech on social media platforms. The entire country faces a difficult task in dealing with the increase of hate speech online, due to the extensive use of social media platforms. Malaysia's cultural variety, which includes Malays, Chinese, Indians, and other indigenous people, is what distinguishes the country. Social media platforms offer a public and private forum which provides a favorable environment for the spread of hate speech where people may air their concerns, biases, and radical views without fear of repercussions.

Hate speech on social media may take many different forms, from offensive memes and discriminatory remarks to disinformation and violent provocation. Hate speech may intensify racial and religious tensions, which can have real-world repercussions including social unrest, strained relationships between communities, and even bloodshed. Other than that, there is also the existence of gender, age hate speech on social media platforms such as Twitter, Facebook, Instagram, Tiktok and other popular platforms. Although many social media platforms

established a few policies and rules to lessen the dangers of hate speech, the varied cultural contexts and huge amount of data have prevented social media from completely and properly detecting hate speech and controlling users. Users will still publish offensive and discriminatory articles on social media as a result of this issue (Velankar et al., 2022).

For example, Malaysia has a thriving football culture, with ardent supporters of several local teams. Rivalries that elicit strong feelings and foster healthy competitiveness are those between Johor Darul Ta'zim (JDT) and Selangor or Kelantan and Terengganu. However, the emergence of social media has accentuated the unfavorable parts of these rivalries, resulting in the spread of hate speech directed at rival clubs and their fans.

Some of the football supporters may express their ideas, celebrate wins, and participate in conversations on social media sites. However, this public digital arena also turns into a haven for hate speech, when people hurl a barrage of disparaging remarks, insults, and threats against rival teams and their supporters. Social media's ability to provide anonymity allows users to attack others without worrying about immediate repercussions, which adds to the feeling of impunity.

The Malaysian government, civil society organizations, and social media platforms have increased efforts to combat the transmission of hate speech in response to the seriousness of this issue. They have been attempting to tighten or adopt laws that prohibit hate speech online. The Malaysian Communications and Multimedia Commission (MCMC) has been aggressively keeping an eye on social media sites and taking action against hate speech propagandists. Although tougher rules can help combat hate speech on social media, they might not be enough to end the problem entirely.

To successfully monitor and implement stricter regulations, there must be sufficient funding, systems, and staff. Governments and law enforcement organizations frequently suffer resource limitations, which makes it difficult to effectively oppose hate speech on a big scale. Therefore, it is important to design an application that could help stop the spread of hate speech on social media in Malaysia and curb hate feelings among Malaysian citizens.

Research Question:

- What are the possible topics that could contribute to social media hate speech in Malaysia?
- What are the impacts that hate speech could bring to Malaysia in terms of mental and physical health?
- What are the effective and efficient solutions to manage the issue of social media hate speech in Malaysia?

Research Objective:

- To identify the possible topics that could contribute to social media hate speech in Malaysia
- To identify the impacts that hate speech could bring to Malaysia in terms of mental and physical health
- To identify the effective and efficient solutions to manage the issue of social media hate speech in Malaysia

Scope:

In this study, there will be 10,000 tweets analyzed for hate speech on two of the social media platforms, Twitter and Facebook. English and Malay will be chosen for the analysis of this study. Our target country in this study is Malaysia.

2.0 RELATED WORKS

Definition

Based on the research from MacAvaney et al. (2019), most social media platforms have their own understanding and guidelines to identify hate speech and manage it. For example, the meaning of the hate speech on Facebook is an outright attack on individuals based on what are known as protected characteristics, including race, ethnicity, national origin, religion, sexual orientation, caste, sex, gender, and gender identity, while the meaning of the hate speech by de Gilbert is a targeted attack against a certain group of people that is purposefully carried out and is driven by the characteristics of that group. (MacAvaney et al., 2019). However, hate speech has become a way for people to express their dissatisfaction through the Internet and social software in modern times. The problem is serious enough to encourage criminal behavior.

To reduce the number of hate crimes, most social media platforms have established a few policies and added some approaches, like hate speech detection. Hate speech detection is a natural language processing (NLP) study to understand, detect and analyze, and monitor the hateful content and sentiment behind the behavior of specific topics (Florio et al., 2020).

Problem of hate speech detection

According to the research from Chakraborty and Masud (2022), we can say that the problem of hate speech detection is subjectivity in language and its context. With the change of the times, the language of human beings is gradually evolving, and the vocabulary of new hate speech is increasing. This has led to the failure of most existing hate speech detection models. Relying on the artificial hate dictionary to capture this change. In addition, there are contextual issues, personal hatred tendencies, and current world views. Even for a person with world knowledge, it is difficult to identify hate speech events.

Research from Yin and Zubiaga (2021) has shown that another problem with hate speech is that the data on it is biased and limited. The undersized data will seriously lead to overfitting when using deep learning and machine learning to query hate speech, and then damage the generalization ability. The biases in data sets are also common. The challenge of developing models that are generalizable across diverse data is further exacerbated by non-random sampling,

subjective annotations, and the various sampling and annotation techniques used across datasets. All current social media datasets for offensive or hate speech material employ boosted (or targeted) sampling with straightforward rules to reduce the impact of a few positive examples on model training. In addition, as seen by various annotation rules, various researches are founded on disparate definitions of "hate speech."

Approach for improving the hate speech detection

According to research conducted by Udanor, C., & Anyanwu, C. C. (2019), special graph-based mining tools will be needed to quickly model the structure of the SN in order to study and assess the value of information from massive amounts of data on the SN. The comparative network analysis tools discussed were Gephi (for interactive network visualization and exploration), Pajek (software for drawing networks, with analytic capabilities, etc.), and IGraph (for creating and manipulating graphs, which is based on the platform, algorithms complexity, execution time, Graph types, features of the graph, and the input file format). For social media hate speech sentiment analysis, the research used cutting-edge NLP approaches by adding part of speech tags to tweets that were gathered and retrieved. The study used unsupervised learning or automatic classification to create the Python Flask sentiment classifier program known as POSA.

Research conducted by Muzakir et al. (2022) proposes identification and classification for improving hate speech detection on social media in Indonesia. They had used the data from communities and figures on Twitter and various methods, such as the SVM algorithm, eXtreme Gradient Boosting (XGBoost), and RandomSearchCV to detect and classify hate speech. They also involve SMOTE (Synthetic Minority Oversampling Technique) to resolve the balanced data. Their research finding is to interpret the results of the accuracy of hate speech detection with and without SMOTE. The result of this research is that using SMOTE can improve the process of detecting hate speech in Indonesia.

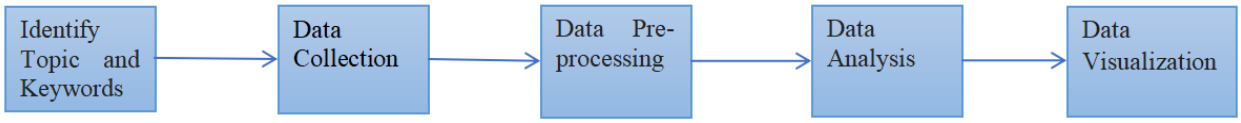
Toliyat et al. (2022) also researched the study about hate speech detection on Twitter in Asia during Covid-19 to improve Asian hate speech detection. They chose a baseline of Americans' hate crimes against Asians on Twitter and 10 million tweets through the Twitter API V-2 as their data. Then, they were using a variety of machine learning and deep learning methods to predict the data, such as K-nearest neighbors (KNN), Support Vector Machine (SVM), Extreme Gradient Boosting (XGBoost),Basic Long-Term Short-Term Memory (LSTM),

Bidirectional LSTM, and Bidirectional Encoder Representations from Transformers (BERT). The final result of the research displayed that logistic regression has the best statistical machine learning performance, while BERT reached the best performance of the deep learning models when detecting hate speech on Asian tweets.

Asiri et al. (2022) have presented the study of introducing the ESGONLP-HSC model, which is based on natural language processing and is used to detect and classify hate speech. This study aims to recognize and classify hate speech on social media. This model, which includes attention-related bidirectional long short-term memory (ABLSTM) and glove-based feature extraction techniques, is used to categorize social media text into three classes: offensive, hateful, and neutral language, then apply the accuracy of the hate speech detection by different machine learning and deep learning models like SVM and TWE. The experimental results show that the ESGONLP-HSC model is superior to the latest method in terms of performance.

3.0 METHODOLOGY

There are a total of five phases in our methodology to perform social network analysis and social media analytics.



Identify Topic and Keywords

The main topic of our analysis is hate speech. A deeper understanding of the topic can be gained by reading articles that are related to it. These articles can be found online using "Google Scholar". After gaining a better insight into the topic, we will identify the hashtags and keywords in order to find posts, tweets or pages related to hate speech on social media. We will get our data mainly from two sites, which are Twitter, a microblogging site, and Facebook, an online social network site.

Data Collection

For data collection, it is difficult to gather and annotate data for the training of the automatic classifiers for detecting hate speech because there is no established definition of what comprises hate speech, it can be challenging to determine if a given text is hate speech. Furthermore, hate speech grows on social media sites, despite the fact that many of them have extremely strict data usage and distribution regulations. As a result, there are limited datasets that can be studied by the general public, with Twitter providing the majority of them. Although the Twitter resources are excellent, their general applicability is constrained. Another problem is that identifying hateful, violent, and insulting text is difficult due to a lack of publicly accessible, tightly controlled datasets. (MacAvaney et al., 2019).

Our proposed system will be able to retrieve data from these social media sites, using BOTSTER. BOTSTER is an online tool that provides functions like extracting data, monitoring resources and automating processes. To extract data, it will scrape web data and import it directly into other apps. For the monitor resources function, BOTSTER can track online resources in real-time and notify users when there are any changes, updates or errors. For automating

processes, it can automate the execution of rules-based repetitive processes at any scale. We have chosen this website to scrape social media data as it does not require the API of those websites before scraping their data. The data collected from Twitter and Facebook using BOTSTER is strictly based on our identified hashtags and keywords related to hate speech, which is done in the first phase.

Data Pre-processing

Next, it is followed by the data pre-processing phase. Before applying any algorithms, we must pre-process the dataset using natural language processing methodologies. Pre-processing is the process of removing noise and components for either sentiment analysis or opinion analytics activities. Lowercasing, tokenizing, and removing punctuation, user mention signs (@), hashtags (#), duplicate characters, URLs, stopwords, and images are all part of this process (Toliyat et. al, 2022). Knowledge Discovery in Databases (KDD) technique is used when the focus is more on meaningful extraction (Vijay et al., 2022).

Data Analysis

In the data analysis phase, which is the forth phase of our methodology, Gephi and Python programming language are used for social network analysis and social media data analytics. Social network analysis is the measurement and mapping of various relational tie distribution pattern properties. It offers both mathematical and graphical analyses of interpersonal ties to reflect the systematic description of networks. Special graph-based mining tools will be needed to quickly model the structure of the social network in order to study and assess the value of information from massive amounts of data on the social network. (Udanor, C., & Anyanwu, C. C., 2019). Gephi is used for social network analysis, as users can quickly configure social network visualization graph using a variety of criteria and features. By using Gephi, we can identify patterns of actor interaction between networks by assessing different centrality metrics, including degree, closeness, and betweenness.

For social media analytics, the Python programming language is used for sentiment analysis on hate speech data collected. Text processing and analysis are done using the Natural Language Toolkit (NLTK). Using the sentiment analysis tool VADER (Valence Aware Dictionary and sEntiment Reasoner), we will get to calculate the proportion of positive, negative, and neutral scores for the hate speech data. We discover that a score of 0 is neutral, a score of less than 0

indicates a negative attitude, and a score of greater than 0 indicates positive polarity. For a lexicon-based approach, a polarity dictionary containing words and phrases annotated by its semantic orientation is necessary (Vijay et al., 2022).

Data Visualization

Finally, the results generated by the system through social network analysis and social media analytics will be visualized. The result of the social network analysis is a social network graph, which users can use to visualize the pattern and other details in it. The graph generated by Gephi will show the detailed connection of the nodes by edges. The relationship between social media posts, which are represented by the nodes is shown. The outcomes of the social media analytics are generated as a pie chart that translates the sentiment scores into numbers and percentages for positive, negative and neutral (Udanor, C., & Anyanwu, C. C., 2019).

4.0 ANALYSIS

Globally, social media is changing how information is transmitted and how cultures are expressed. The number of people using social media to express themselves, communicate, share information, and connect with friends has greatly expanded globally in recent years. However, the portability and anonymity of online media allow internet users to spread offensive information from behind the screen. Twitter and Facebook are two examples of large social media platforms that are under criticism for failing to take action against hate speech (HS) on their platforms, although several algorithms were implemented on their sites to filter out some offensive tweets or posts (Asiri et al., 2022).

Even for humans, it can be challenging to distinguish the different types of text that make up hate speech. In addition , manual hate speech judgment takes a lot of time and also creates a personal sense of hate speech composition. Due to the vast amounts of Internet material, an automatic HS identification and prediction approach should guarantee that the Hate-Speech is strictly limited on mass social media platforms, especially Twitter and Instagram.

Therefore, in order to curb and reduce hate speech on social media platforms, an SocMed application named “Guard Hate Social Media” is designed to handle Malaysia's rising problem with hate speech on social media. There are two main functions in the proposed application, which are generating social network graphs and performing social media analytics on the collected dataset based on our specific keyword related to our topic, which is hate speech in Malaysia.

In order to freely express themselves, meet new people, and exchange ideas or hobbies, the social network links individuals with friends and family in a social circle. The social network is also an expression of social relationships and connections made between individuals known as nodes (i.e., objects, people, or other members of the network), which are connected by links, edges, or ties that are based on various types of specific interdependencies, such as dislike, friendship, mention and retweet (Udanor, C., & Anyanwu, C. C., 2019).

On the other hand, social network analysis (SNA) sees social connections as nodes and ties in a network theory. SNA is a technique for measuring, representing, and analyzing social structure that aims to forecast how relationships between social entities will be structured as well as how that structure will affect other social phenomena. A hierarchy of communities within the main issue can be created by using SNA to investigate a network topology through the connections that are formed within it. Additionally, SN like Twitter and Facebook enable relationship tracking in situations where knowledge is not shared. In other words, a node follows another node, and the quantity of followers somewhat determines a node's popularity inside the network, but it cannot determine if this popularity is good or bad (Udanor, C., & Anyanwu, C. C., 2019).

As a result of the increased usage of social network graphs as a strong and useful tool for NLP applications, performance for tasks like text classification, sentiment analysis, and author attribute recognition has significantly improved. The idea of utilizing this kind of information is best explained by the idea of homophily, or the phenomenon that people like to associate more with those that seem similar both in real life and online. Similarity can be determined here depending on a variety of factors, including geography, age, language, etc. Leveraging community interaction, it is found that if we have knowledge about members of a group that is identified by some similarity metric, we can infer details about an individual based on the group to which they belong (Chowdhury et al., 2019).

A social network graph is created using Gephi. Gephi is the best software for visualization and exploration for all types of graphs and networks. The social network graph may represent the relationships between users, the content they post, and the interactions between users and reported content. It can help identify patterns of hate speech and track the spread of hateful content.

Besides, the second main function proposed is social media analytics. Social media analytics is defined as using advanced informatics tools and analytics techniques to collect, monitor, and analyze social media data to extract useful patterns and intelligence (Hassani, A., & Mosconi, E. , 2022). The suggested model must be able to handle the continual growth and evolution of hate speech, regardless of its expression, audience, or source (Teh, P. L., & Cheng, C. B., 2020).

Hence, the proposed model is set to have automatic approaches for hate speech detection and perform sentiment analysis on hate speech detected. The majority of social media sites have set up user guidelines that restrict hate speech, but implementing these guidelines requires a lot of human work because each report must be reviewed. The reviewing process might be sped up using automatic tools and methods (MacAvaney et al., 2019).

Sentiment analysis is very helpful for identifying hate speech on social media. Depending on the language used in a given post on a particular issue, sentiment might be interpreted as being positive, negative, or neutral. It is important to comprehend the words and categorize sentiments into their categories in order to do sentiment analysis. Sentiment analysis lies between NLP and natural language understanding (Udanor, C., & Anyanwu, C. C., 2019).

5.0 EXPECTED RESULT & EXPERIMENT

As the network evolves, more people begin to use social media platforms such as Facebook, Instagram, and Twitter. Nowadays, people have more freedom to express themselves, and they like using social media to say whatever they want. One of the things we need to take seriously from that is hate speech. A message or communication that disparages a person or group on the basis of their religion, race, caste, sexual orientation, gender, nationality, or other traits is known as hate speech.

A public statement that incites, promotes, or justifies hatred, discrimination, or hostility towards a particular group based on their race or ethnicity, beliefs or faith, abilities or disabilities, age, sexual orientation, or gender identity is referred to as using "malicious language", "hate speech," or "flaming comments," and requires government regulation because it fosters, incites, and encourages prejudice, violence, and discrimination against certain people or groups, hate speech is repugnant (Azman & Zamri, 2022).

As a result, we can draw the conclusion that hate speech is a form of vulgar, obscene, abusive, and abusive language that frequently expresses hatred and sometimes even uses blasphemous words. Online hate speech can hurt victims by causing post-traumatic stress disorder (PTSD), which includes psychological and pathophysiological symptoms such as panic, fear, anxiety, nightmares, and disdain. Prior to this, a great deal of scholarship was devoted to understanding, identifying, and analyzing hate speech on social media (Azman & Zamri, 2022).

Hate speech can be defined broadly to include insults, discrimination, dehumanization, demonization, and incitement to violence. It refers to emotions and words as well as meanings that can evoke emotions in someone who hates a particular object. To the untrained eye, hate speech is made up of two words that are very common: hatred and speech. Hate speech and hate messaging include everything and everything that criticizes other people. Sarcasm, insults, and other harsh language can appear to be "innocently" negative. Words like "babi" or "anjing" are extremely offensive in Malaysian cultures, yet "pig" or "dog" are not offensive in American culture. In addition, calling people by certain tribe names, including "batak," "sakai," and "keling," can be hurtful and have a bad impact on other users (Azman & Zamri, 2022).

Ties are links between people that allow them to share information, knowledge, and experience. Depending on how much the two nodes connect with each other, these links can be strong, weak, or not there at all. In social networking, it's important to know the difference between strong and weak ties. Both strong and weak ties are important for building a group, but in different ways. Someone with a strong tie (close friends) is someone who is in a group of close friends and family. Weak ties (familiarity) are bridges between two different groups (SN). Absent or hidden ties are connections that should be there but aren't (Udanor & Anyanwu, 2019).

We will first determine the nodes (Twitter and Facebook postings) and edges (links) between them to build the social network graph. We will perform sentiment analysis on every Twitter and Facebook postings and comments on the constructed graph to gain insight regarding hate speech networks. To comprehend the dynamics of debates involving hate speech, we shall examine the relationships between posts with other posts and comments with other comments. Recognise patterns of activity, such as responses, retweets, or comments that serve to amp up hate speech. We will then use sentiment analysis tools to determine if postings and comments are positive, negative, or neutral. This can assist detect really negative or inflammatory content and offer information about the network's overall mood regarding hate speech.

After that, we will utilize diameter in Gephi to perform community detection algorithms to identify clusters or groups within the graph that share similar characteristics or engage in similar hate speech discussions. Finally, we will create visualizations of the social network graph and its various properties to aid in understanding and communicating the findings. Visualizations can highlight connections, community structures, and important nodes, making it easier to interpret the hate speech network.

We searched the hashtag #HarimauMalaya on Twitter and found a post by New Straits Times that said, “The Coldplay concert at the National Stadium on Nov 22 may result in only limited seats for football fans if Harimau Malaya decide to play their 2027 Asian Cup qualifying match at the same venue a day before” (Times, N. S., 2023). One user comments inside this post about "PSM memang bangang." The Football Association of Malaysia is foolish, according to this comment. The word "bangang" denotes someone who is dumb, which can be classified as hate speech.

When we searched for the keyword #HarimauMalaya, we also discovered a post by Astro Awani from Twitter. The post says “FAM tidak akan membenarkan konsert Coldplay mengganggu perlawanan Kelayakan Piala Asia 2027 yang akan diadakan sehari sebelum itu” (AWANI, A., 2023) As a result, the Football Association of Malaysia will not host the Coldplay concert the day before their 2027 Asian Cup qualification game. One user commented on this post saying “Cancel je. Sbb aku xdpt ticket. Babi punya scalpers”. In addition to being offensive to Muslims, the word "babi" is used in this comment to also mean unclean, unhygienic, and other things. This caused the word to become insulting in Malaysia.

6.0 CONCLUSION

Hate speech on social media is a serious problem for Malaysia, a multicultural country with a lively online community. Hate speech threatens societal cohesion by promoting tension, hostility, and division among various populations. Even though attempts have been made to solve this problem, it is still a complicated and ongoing issue.

Research is crucial to comprehending the underlying causes, effects, and successful treatments in order to successfully counteract hate speech. By doing this study, we can identify solutions to manage social media hate speech in Malaysia. By solving this issue, it can offer many advantages to Malaysian citizens and netizens. They could have greater exposure to the latest information and technology in a safe manner.

In addition, combating hate speech on social media promotes a more unified and inclusive society. People from many racial, religious, and cultural backgrounds might feel safer and more respected online if hate speech is curbed on social media platforms. As a result, societal harmony is increased, and intergroup interactions are improved. Solving social media hate speech could also increase Malaysia's reputation. It reflects the country's dedication to social advancement, inclusiveness, and human rights. This can improve a country's tourism, promote cross-border cooperation in a variety of sectors, and even improve political stability.

The elimination of hate speech from the internet promotes productive conversation, courteous debate, and pleasant encounters. It encourages a society that values empathy, comprehension, and respect for one another by enabling people to express their ideas and have productive dialogues.

In a nutshell, it is essential to develop a digital environment in Malaysia where respect, empathy, and conversation triumph over hate speech in order to appreciate its variety and pave the way for a more peaceful and inclusive society. Therefore, the application "Guard Hate Social Media" is aimed at reducing social media hate speech in Malaysia.

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